Digital Adoption, Business Performance, and Financial Literacy in Ultra-Micro, Micro, and Small Enterprises in Indonesia

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ABSTRACT

This study investigates the factors affecting digital adoption by ultra-micro, micro, and small enterprises (UMSEs) based on a survey of 5,035 UMSEs in 17 major provinces in Indonesia. We use the survey data to construct a digital adoption index to evaluate the variations in regional digital adoption. The results demonstrate that owner demographic characteristics, firm-specific factors, business environment, connectivity infrastructure quality, and culture are associated with disparities in digital adoption by UMSEs. Our findings also reveal a positive and significant correlation between digital adoption on business performance (sales growth). In addition, we found strong evidence of the impact of digital adoption on UMSE owners' level of financial literacy. The latter result suggests that improving digital adoption among UMSEs could enhance their financial literacy. Overall, these findings confirm the vital role of digital transformation for UMSEs to achieve growth and competitiveness in the global market, which requires robust support from policymakers.

Keywords: Digital Adoption, Ultra-Micro, Micro and Small Enterprises, Firm Performance, Financial Literacy.
JEL Classification: L25 O33 R11

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1 Introduction

Microbusinesses, notably ultra-micro, micro, and small enterprises (UMSEs) have a significant role in economies, particularly in developing countries like Indonesia.[1] Typically, micro businesses are a source of income generation and job creation for many households (Mead & Liedholm, 1998; Trinugroho et al., 2022). UMSEs can also serve as an informal social safety net by providing alternative jobs following extraordinary disruptive circumstances (Resosudarmo et al., 2012; Falentina et al., 2021). However, some issues remain regarding the productivity and long-term viability of such enterprises, although the Indonesian government has implemented some affirmative policies and facilities (Falentina et al., 2021; Trinugroho et al., 2022).

In the last two decades, digital transformation has transformed all sectors due to the penetration of information and communications technology (ICT), most notably via the internet. Many UMSEs in Indonesia have also adopted digital technology for conducting business activities; for instance, e-commerce, online marketing, digital payments, and digital banking services (e.g., Ridhwan et al., 2023). Nevertheless, the level of adoption is uneven across UMSEs, particularly in developing countries. Previous studies have highlighted the benefits of digital adoption by micro, small, and medium-sized enterprises. Examining European small- and medium-sized enterprises (SMEs), Skare et al. (2023) provided evidence that digital transformation in SMEs significantly impacts business activities, competitiveness, input costs, access to finance, and skilled labor demand. In the context of Indonesia, adopting digital technology can be essential for micro and small enterprises, particularly during the COVID-19 pandemic (Trinugroho et al., 2022; Ridhwan et al., 2023).

Accordingly, this study examines UMSEs' digital adoption and the factors that could explain the variations in adoption across regions in Indonesia. Based on field survey data, we build a comprehensive index to measure each enterprise's level of digital transformation, identifying five UMSE business processes, which include e-procurement, point of sale (POS), e-marketing, e-commerce, and digital payment. Previous studies have only employed separate indicators (using dummy variables) to identify digital adoption, and these five considerations could potentially be transformed into a *digital adoption index*. Our index identifies

[1] We refer to Ridhwan et al. (2023) for the UMSE construct based on the Indonesian government law *PP No.7 Tahun 2021.*

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whether a firm has adopted digital processes and considers the usage dimension to measure the level of adoption more precisely.

This paper also contributes to the literature on micro businesses' digitalization by empirically assessing the determinants of digital business adoption by ultra-micro,[2] micro, small, and medium-sized enterprises in Indonesia in 2023. Another contribution of this study is our examination of whether digital adoption affects UMSEs' financial literacy. Our dataset is sourced from the Bank Indonesia Institute Survey of Ultra-Micro, Micro, and Small Enterprises, which is not publicly available.

2 Related Literature

Previous research has explored digital transformation in business entities extensively. Santoso et al. (2020) contended that internet penetration and the ubiquity of smartphones are the most significant drivers of digital adoption from the supply and demand sides. In a survey measuring readiness for digital financial services, Trinugroho et al. (2017) found that higher quality infrastructure increases business readiness to use digital financial services in Indonesia. From a different perspective, management and behavioral science scholars have primarily employed the technology acceptance model (TAM) to explain the determinants of digital adoption by individuals or organizations. For instance, in a literature review, Kajol et al. (2022) identified 15 behavioral factors derived from TAM that motivate the adoption of digital financial transactions and overall digital adoption. Furthermore, the continuous evolution of digital innovation in finance (fintech) is also driven by the development of e-commerce, which requires support from the financial services sector for several functions, particularly payment transactions. Wirdiyanti et al. (2022) empirically confirmed this hypothesis by providing evidence that engagement in e-commerce by micro, small, and medium-sized enterprises could lead to better firm performance. Moreover, financial inclusion also improved following the adoption of e-commerce.

The benefits of digital adoption, particularly by small businesses, have also been explored in previous research. For instance, examining micro and small enterprises (MSEs) in South Africa, Gaglio et al. (2022) determined that using

[2] We introduce an ultra-micro enterprise category in this study and present the justification for this inclusion in section 3.1.

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social media and business mobile phones for surfing the internet has a positive effect on innovation, and innovation conditional on the use of such technologies has a positive effect on labor productivity. Trinugroho et al. (2022) found that digital adoption, measured by three indicators of digital payment, POS, and online marketing, was positively associated with small businesses' performance during the COVID-19 pandemic. In contrast, Buck et al. (2023) argued that data security and privacy issues are the most striking potential dark side of digital adoption.

Some studies have revealed that digital adoption, particularly that of MSEs, could enhance the level of financial inclusion. Financial inclusion refers to all initiatives that make formal financial services accessible and affordable, particularly for underserved populations (Omar and Inaba, 2020) across three considerations of financial services' access, usage, and quality. Digital adoption by small businesses could improve the financial inclusion of MSEs as mobile digital payments can provide the structured information related to daily transactions that is needed when applying for loans. Avom et al. (2023) demonstrated that the adoption of mobile money in 50 African countries promotes financial inclusion in a range of 12%–14%. Similarly, using household surveys in China at the micro level, Yang and Zhang (2022) document that fintech adoption significantly affects household consumption and reduces consumption inequality.

Digital adoption is also positively associated with financial literacy, meaning that using digital platforms could improve individual financial literacy. Using country-level data, Lo Prete (2022) found that higher financial literacy is correlated with the use of digital payments. Likewise, Yang et al. (2023) provided evidence that financial literacy increases the use of digital finance in China, such as mobile payments, online borrowing, and online financial products.

3 Research Methods

3.1 Data

We conducted a field survey of UMSEs to achieve the objectives of this study, obtaining responses from 5,035 firms via in-person and online surveys because it is essential to include numerous observations to mitigate potential sample selection bias. Regional economic development data are retrieved from the Indonesian Statistics Office Bureau.

We employed multistage sampling for the survey, combining quota (province level) and convenience sampling. However, we also proportionately balanced

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factors of 1) business sector, 2) rural/urban, and 3) firm size when organizing the observations. The survey covers 17 major provinces in Indonesia, representing around 80% of the Indonesian economy.

- Sumatera island: North Sumatera, West Sumatera, Riau, Lampung, and South Sumatera;
- Java–Bali island: DKI Jakarta, Banten, West Java, Central Java, DI Yogyakarta, East Java, Bali;
- Kalimatan island: West Kalimantan, East Kalimantan, and North Kalimantan;
- Sulawesi island: South Sulawesi and Central Sulawesi.

UMSEs' scale is classified based on revenue in the last financial year and adhering to the latest regulations set by the Indonesian government, as stipulated in *PP No. 7 Tahun 2021*.[3]

Referencing Ridhwan et al. (2023), the classification of the businesses in our study is divided into ultra-micro, micro, small, and medium-sized enterprises, as shown in Table 1. Ultra-micro enterprises are defined as those with an annual revenue not exceeding Rp 250 million, and the definitions of micro, small, and medium-sized enterprises align with current Indonesian regulations.

Table 1 UMSE criteria based on annual revenue

Type of Enterprise	Revenue Criteria (Indonesian Rupiah / Rp)	Reference
Ultra-micro enterprise	Maximum 250 million	Novelty of this study
Micro enterprise	>250 million–2 billion	
Small enterprise	>2 billion–15 billion	
Medium-sized enterprise	>15 billion–50 billion	Indonesian government law *(PP No.7 Tahun 2021)*

3.2 Digital Adoption Index

We construct an index to measure UMSEs' digital adoption, ranging from 0 (not adopted at all) to 1 (fully adopted digital). As explained previously, we include five business processes of e-procurement, POS, e-marketing, e-

[3] See https://jdih.setkab.go.id/PUUdoc/176384/PP_Nomor_7_Tahun_2021.pdf for further details.

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commerce, and digital payment. These five processes could potentially be digitally transformed; thus, we consider the channels through which UMSEs adopt digital technology based on these considerations (Table 2).

Table 2. Measurement of Digital Adoption Index[4]

No	Subindex	Indicator	Measure
1	E-Procurement	Does your business conduct procurement (purchase of goods) online?	Dummy
		Does your business make online payments to suppliers?	Dummy
		Of all your needs for procuring production materials or business inventory, what percentage do you procure online?	Percentage
2	Point of Sale	Do you use a digital cashier/point of sale (POS) application in your business activities?	Dummy
		What features do you use in your digital cashier (POS) application?	Percentage
3	E-Marketing	Do you market your products online?	Dummy
		What social media do you use?	Percentage
		Do you regularly update information about your business products/services on social media?	Dummy
		Have you ever used paid advertising on social media to market your products?	Dummy
		Does your business have a website?	Dummy
		Do you regularly update information about your business products/services on your website?	Dummy
		Of your total marketing costs, what percentage is allocated to online marketing?	Percentage
4	Digital Payment	What digital/non-cash payment alternatives do you provide?	Percentage
		If you provide QRIS[5], what is the percentage of payments via QRIS in your total business income?	Percentage
		What percentage of your total business income is received through digital payments?	Percentage
5	E-Commerce	Do you use e-commerce to sell your products?	Dummy
		Please mention the e-commerce used to sell your products.	Percentage
		What percentage of your total sales income is obtained through e-commerce sales?	Percentage

[4] We perform Cronbach's alpha (a) to check the reliability of each subindex and aggregate index. The results show that all Cronbach's a scores are higher than 0.6, indicating that our index and subindices are reliable (Malhotra & Dash, 2010).

[5] QRIS is Quick Response Code Indonesian Standard which is a national QR code standard to facilitate QR code payments in Indonesia.

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E-procurement refers to the procurement of goods that is conducted by UMSEs using technology and online media. POS is UMSEs' use of an application to support transaction recording and bookkeeping that could provide sales, product prices, customer data, inventory, income, expenses, and profit information. Digital payment refers to non-cash payment methods. E-marketing involves advertising products through online media such as social media and websites. E-commerce refers to sales through e-commerce platforms.

Previous studies have employed dummy variables to identify digital adoption (e.g. Trinugroho et al., 2022; Wirdiyanti et al. 2022). The digital adoption index in this study not only identifies whether a firm has adopted digital technology but also considers the dimensions of use to measure the level of adoption more precisely. Table 2 provides the detailed measures (indicators) of our digital adoption index. Each subindex (e-procurement, POS, digital payment, e-marketing, and e-commerce) consists of two dimensions covering adoption and usage.

3.3 Variable Definitions

As previously described, we conduct several empirical estimations to investigate the determinants of digital adoption and examine the effect of digital adoption on firm performance, financial inclusion, and financial literacy. Table 3 provides the variable descriptions and measurement details.

Table 3. Variable Descriptions and Measures

Variable	Notation	Indicator	Description	Source
Digital Adoption	DIGI	Digital Adoption Index	This index includes five aspects of digital adoption by UMSEs (e-marketing, e-commerce, digital payment, point of sale (POS), and e-procurement). It covers whether firms adopt digital channels and considers the extent to which the technology is used.	Research Survey
Firm Performance	PERF	Sales growth	Annual sales growth (percentage).	Research Survey
Finance Inclusion	INCL	Finance Inclusion Index	This index includes bank account usage and e-payment services provision.	Research Survey
Financial Literacy	LIT	Financial Literacy Index	This index includes seven basic financial problem questions.	Research Survey

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Variable	Notation	Indicator	Description	Source
Firm-Specific Factors	FIRM	Firm Age	Calculated as 2023 – the year when the was firm established.	Research Survey
		Sector	A set of dummy variables including six sectors: 1. Agriculture 2. Transportation 3. Information and communications technology (ICT) 4. Industry 5. Trade 6. Other services	Research Survey
		Firm Size	1. Ultra-micro (1: annual revenue less than IDR250 million, 0 otherwise). 2. Micro (1: annual revenue less than IDR2 billion and more than IDR250 million, 0 otherwise). 3. Small (1: annual revenue more than IDR2 billion and less than IDR15 billion, 0 otherwise).	Research Survey
		Business License	Dummy variable (1: has a license, 0 otherwise).	Research Survey
		Home-based location	Categorical variable equals 2 if the business site is the same as the household residence, 1 otherwise.	Research Survey
Owner Demographic Characteristics	OWNR	Age	Calculated as 2023 – the year when the owner was born.	Research Survey
		Education	Scale 1–7: 1. No schooling 2. Elementary school 3. Junior school 4. Senior school 5. Diploma I/III 6. Undergraduate/Diploma IV 7. Graduate/Doctoral	Research Survey
		Gender	A dummy variable (1: male, 0: female).	Research Survey
Business Environment	BUS	Digital Competition	A dummy variable for the availability of online payment for other surrounding firms. 1 means yes, 0 otherwise.	Research Survey

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Variable	Notation	Indicator	Description	Source
		Banking Relationship	A dummy variable (1: borrow money from bank, 0 otherwise).	Research Survey
ICT Infrastructure	INFRA	Smartphone Society	Scale 1–5 (measuring the extent to which the surrounding community where the UMSEs is located uses smartphones in their activities).	Research Survey
		Technology awareness	Scale 1–5	Research Survey
Culture	CULT	Trust to digital	Scale 1–5 (measuring the extent to which people trust digital processes in general).	Research Survey
Demographic and Community Characteristics	DEM	Java	A dummy variable regarding an island location (1: Java, 0: non-Java).	Research Survey
	ELE	Elevation	The average elevation of districts above the sea surface where UMSEs operate.	USGS
	NET	Internet user	The average internet users in Indonesian districts.	Susenas

3.4 Empirical Methods

We employ the following models to address the research questions.

1. *Factors affecting digital adoption*

$$Index_i = X_i.\gamma + \theta_k + \theta_l + \varepsilon_i \quad (1)$$

where y_i represents the dependent variable of each digital index for observation i. X_i is a matrix of other owners and UMSEs' characteristics for observation i, with γ as the vector of coefficients. θ_k and θ_l are sectoral dummy variables and district characteristics variables for categories k and l. Finally, ε_i is the error term for observation i.

2. *Impact of digital adoption on firm performance*

$$SalesGrowth_i = \beta Index_i + X_i.\gamma + \theta_k + \theta_l + \varepsilon_i \quad (2)$$

where $SalesGrowth_i$ represents is outcome variable for observation i, and β is the coefficient for the independent variable $Index_i$, which indicates the effect of digitalization on the dependent variable. X_i is a matrix of other owners'

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characteristics and UMSEs' characteristics for observation i, with γ as the vector of coefficients. θ_k and θ_l represent sectoral dummy variables and district characteristics variables for categories k and l. ε_i is the error term for observation i.

3. Impact of digital adoption on firm-level financial Literacy

$$FinLit_i = \beta Index_i + X_i.\gamma + \theta_k + \theta_l + \varepsilon_i \quad (3)$$

where $FintLit_i$ represents the Financial Literacy Index for observation i, and β is the coefficient for the independent variable $Index_i$, which indicates the effect of digitalization on the dependent variable. X_i is a matrix of other owner's and UMSEs' characteristics for observation i, with γ as the vector of coefficients. θ_k and θ_l represent sectoral dummy variables and district characteristics variables for categories k and l, and ε_i is the error term for observation i.

We acknowledge the potential endogeneity in Equations 2 and 3 as some may argue that higher digital indices are related to some unobserved variables in the error. Therefore, to isolate this issue, we employ two-stage least squares (2SLS) to estimate the models. The instrumental variables (IVs) are digital competition and signal strength for Equations 2 and 3, respectively. The following is the first-stage equation of our IV estimation:

4. First-stage IV estimates for predicting digital adoption indices in business performance outcome

$$Index_i = \beta DigCompetition_i + X_i.\gamma + \theta_k + \theta_l + \varepsilon_i \quad (4)$$

where *DigCompetition* is the availability of online payments for the surrounding UMSEs of enterprise i.

5. First-stage IV estimates for predicting digital adoption indices in financial literacy outcome

$$Index_i = \beta SignalStrenght_i + X_i.\gamma + \theta_k + \theta_l + \varepsilon_i \quad (5)$$

where *signal strength* is the signal speed of UMSEs' location at enterprise i.

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4 Results

4.1 Variables' Descriptive Statistics

Table 4 presents a comprehensive overview of the various economic and demographic characteristics of 5,035 respondents, revealing modest sales growth among businesses with an average of around 19%. The digital adoption indices show varying levels of adoption, with e-procurement reaching a relatively higher mean (29.2%) than others and the aggregate index of the five indices denoting a mean of 20.3%. The data also reflect a combination of nascent and long-established firms with a diverse owner demographic in terms of age, education, and gender. Substantial use of smartphones and the internet is also evident, indicating that people are well-intertwined with technology use. Many of the respondents are located on Java Island, a place known for its tech-savvy development.

Table 4. Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Sales Growth	5,035	0.193	0.152	0.000	1.000
Fin Literacy Index	5,035	0.719	0.202	0.000	1.000
Fin Inclusion Index	5,035	0.741	0.314	0.000	1.000
Aggregate Index	5,035	0.203	0.176	0.000	0.824
Index E-commerce	5,035	0.162	0.254	0.000	0.944
Index E-Procurement	5,035	0.292	0.371	0.000	1.000
Index POS	5,035	0.100	0.257	0.000	1.000
Index Digital Payment	5,035	0.203	0.175	0.000	0.889
Index E-Marketing	5,035	0.260	0.202	0.000	0.964
Firm Age	5,035	6.810	5.837	0.000	75.000
Business License	5,035	0.384	0.486	0.000	1.000
Home-Based Location	5,035	0.353	0.478	0.000	1.000
Owners Age	5,035	34.725	11.715	16.000	87.000
Dummy Male	5,035	0.373	0.484	0.000	1.000
Education	5,035	4.606	1.080	1.000	7.000
Digital Competition	5,035	0.702	0.457	0.000	1.000
Banking Relationship	5,035	0.171	0.376	0.000	1.000
Internet Connectivity	5,035	3.906	0.673	1.000	5.000
Smartphone Society	5,035	0.952	0.215	0.000	1.000
Technology Awareness	5,035	0.796	0.164	0.200	1.000
Trust to Digital	5,035	0.760	0.121	0.200	1.000
Internet Use	5,017	0.667	0.094	0.305	0.840
Elevation	5,034	271.970	252.054	1.014	1509.357
Log Sales	5,035	7.971	0.495	6.556	10.079
Log GDP	5,035	4.783	0.206	4.569	5.475
Dummy Java (the rest of regions = 0)	5,035	0.773	0.419	0.000	1.000

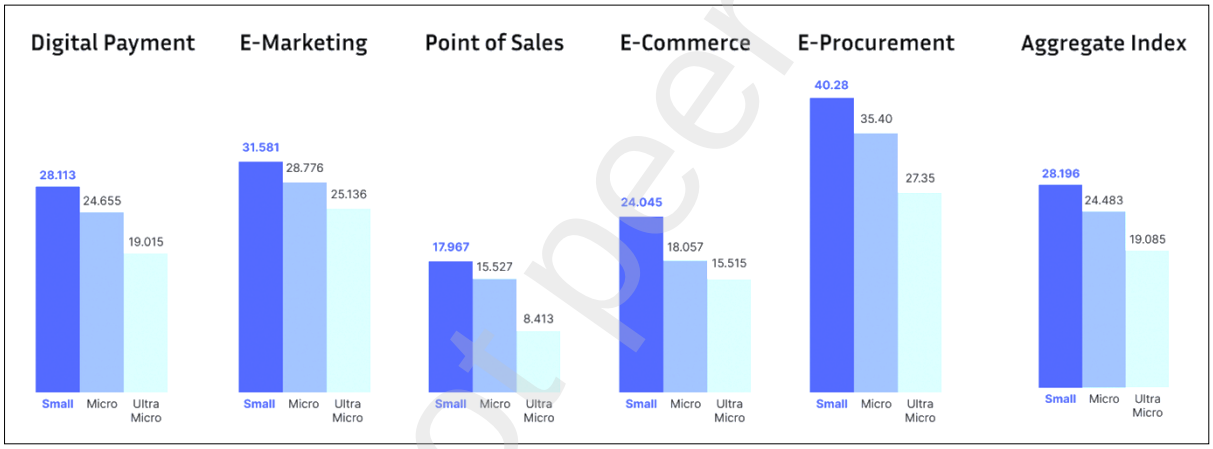
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As noted previously, we group each UMSE size based on revenue. Ultra-micro includes businesses with less than IDR250 million, micro businesses are businesses earning between IDR250 million and IDR2 billion, and businesses with revenue between IDR2 billion and IDR15 billion are small-scale businesses.

Figure 1 illustrates the survey result regarding the digital adoption index by firm type, revealing a notable trend of larger businesses appearing using digital technology more than their cohorts, indicating that bigger businesses are more engaged and integrated with digital business processes than their smaller counterparts. The lower index of ultra-micro businesses could signify the importance of more support in the digital adoption process.

Figure 1. Average digital adoption index by company scale



Looking more deeply into the average index by each sector, health, education, and other services scores the highest average index by 24.05 (Figure 2). This sector could be more robust in terms of measurement criteria, and is possibly driven by high levels of digital transformation and innovation. In contrast, agriculture, livestock, forestry, and fishery have the lowest score of 13.86, suggesting that these primary industries may face challenges that hinder digital performance development. Nevertheless, each sector's index could be influenced

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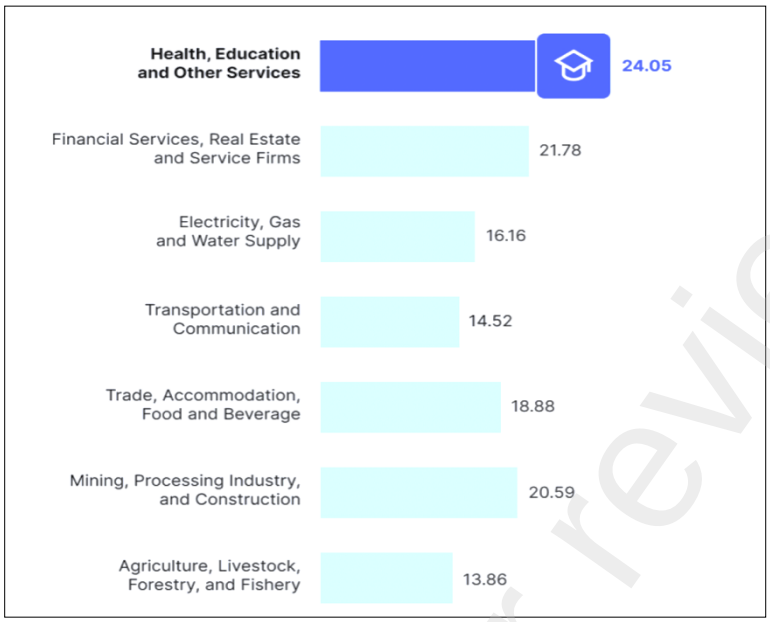
by a range of factors, including market demand, regulatory environments, and investments in technology.[6]

[6] For the remaining descriptive analysis, please visit the following link: https://rb.gy/vmcbir

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Figure 2. Average index by sector



4.2 Empirical Results

As explained previously, our study empirically investigates the determinants of digital adoption by UMSEs in Indonesia, while also testing the effect of digital adoption on firm performance, financial inclusion, and financial literacy. To examine the determinants of digital adoption, we include several considerations, including firm-specific factors, owner demographic characteristics, business environment, quality of ICT infrastructure, and culture that potentially determine digital adoption by MSEs. Our dependent variable in the first empirical model is digital adoption, as explained in section 3. We use sales growth as a proxy for firm performance and build indices to measure the two variables of financial inclusion and financial literacy.

Table 5 presents the regression results of the factors affecting digital adoption based on Equation 1. Table 6 provides the empirical results regarding the effect of digital adoption on firm performance (sales growth). Table 7 exhibits more robust regression results on the effect of digital adoption on firm performance by instrumenting the index with a digital competition variable. Tables 8 and 9 present regression results on the effect of digital adoption on financial literacy using signal strength as the IV.

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Table 5. Factors Affecting Digital Adoption

	(1) Aggregate Index	(2) Index E-commerce	(3) Index E-Procurement	(4) Index POS	(5) Index Digital Payment	(6) Index E-Marketing
Firm Age	−0.00191***	−0.00141*	−0.00373***	−0.000885	−0.000614	−0.00290***
	(−4.83)	(−2.22)	(−4.14)	(−1.31)	(−1.50)	(−6.38)
Business License	0.0114*	−0.00740	0.0210*	0.0256**	−0.00977*	0.0275***
	(2.48)	(−1.00)	(2.00)	(3.27)	(−2.06)	(5.21)
Home-Based Location	0.0255***	0.0210**	0.0244*	0.0450***	0.0236***	0.0134**
	(5.70)	(2.92)	(2.39)	(5.90)	(5.11)	(2.61)
Owners Age	−0.00212***	−0.00273***	−0.00247***	−0.00183***	−0.00165***	−0.00194***
	(−10.69)	(−8.56)	(−5.45)	(−5.40)	(−8.03)	(−8.47)
Dummy Male	−0.0164***	−0.0275***	−0.0405***	0.0226**	−0.0132**	−0.0235***
	(−3.75)	(−3.91)	(−4.05)	(3.02)	(−2.92)	(−4.66)
Education	0.0199***	0.0232***	0.0269***	0.0170***	0.0148***	0.0175***
	(9.87)	(7.18)	(5.86)	(4.94)	(7.12)	(7.54)
Digital Competition	0.130***	0.141***	0.197***	0.0354***	0.137***	0.138***
	(26.44)	(17.88)	(17.64)	(4.24)	(27.01)	(24.43)
Banking Relationship	0.0196***	0.00629	0.0284*	0.0436***	0.0125*	0.00726
	(3.44)	(0.69)	(2.18)	(4.48)	(2.11)	(1.11)
Smartphone Society	0.0248*	0.0345*	0.0322	0.00964	0.00731	0.0403***
	(2.52)	(2.18)	(1.43)	(0.57)	(0.72)	(3.56)
Technology Awareness	0.103***	0.0433*	0.204***	0.0909***	0.108***	0.0694***
	(7.66)	(2.00)	(6.64)	(3.97)	(7.76)	(4.49)
Trust to Digital	0.169***	0.0186	0.320***	0.0770*	0.142***	0.288***
	(9.35)	(0.64)	(7.76)	(2.50)	(7.59)	(13.85)
_cons	−0.171***	−0.0406	−0.357***	−0.166***	−0.118***	−0.172***
	(−7.56)	(−1.12)	(−6.94)	(−4.32)	(−5.04)	(−6.61)
Industry	YES	YES	YES	YES	YES	YES
N	5,035	5,035	5,035	5,035	5,035	5,035

Note: t statistics are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 5 reveals that the business environment, which is measured by competitive pressure, is positively associated with digital adoption. Moreover, technological awareness and trust in digital processes significantly and positively affect digital adoption. Notably, digital adoption by UMSEs is higher in communities where smartphones are widely used. Furthermore, UMSEs with commercial bank relationships are more likely to adopt digital technology.

Owner demographic characteristics are relevant for explaining disparities in the adoption and use of digital technologies. In particular, the elderly are less likely to use digital technologies for their business. We also find that digital technology is more likely to be adopted by more educated owners. Moreover, digital technology is employed more by UMSEs with female owners.

Regarding firm-specific factors, older firms are negatively associated with digital adoption, except for POS and digital payment. The separation of business and owner house has a positive and significant effect on all digital adoption considerations. Similarly, having a business license appears to increase digital adoption by UMSEs.

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Table 6. Digital Adoption and Business Performance (Sales Growth)

	(1) Sales growth	(2) Sales growth	(3) Sales growth	(4) Sales growth	(5) Sales growth	(6) Sales growth
Aggregate Index	0.137***					
	(0.0133)					
Index E-Commerce		0.0336***				
		(0.00867)				
Index E-Procurement			0.0572***			
			(0.00603)			
Index POS				0.00930		
				(0.00840)		
Index Digital Payment					0.150***	
					(0.0128)	
Index E-Marketing						0.107***
						(0.0113)
Transportaion	1.129	1.425	1.459	1.383	1.188	1.376
	(1.077)	(1.087)	(1.079)	(1.091)	(1.074)	(1.079)
ICT	−4.417*	−4.278*	−4.384*	−4.415*	−4.990**	−4.014
	(2.501)	(2.524)	(2.505)	(2.528)	(2.494)	(2.506)
Industry	1.694	2.115*	2.026*	2.195*	2.064*	1.472
	(1.150)	(1.160)	(1.151)	(1.161)	(1.146)	(1.154)
Trade	−0.941	−0.472	−0.888	−0.328	−0.692	−0.446
	(1.114)	(1.123)	(1.116)	(1.124)	(1.109)	(1.114)
Other Services	2.774**	3.601***	2.822***	3.649***	3.108***	3.068***
	(1.080)	(1.086)	(1.082)	(1.088)	(1.074)	(1.080)
Elementary School	6.349	6.966	6.639	7.043	6.260	6.470
	(4.651)	(4.693)	(4.659)	(4.700)	(4.638)	(4.660)
Junior High School	3.021	3.447	3.191	3.513	2.920	3.067
	(4.518)	(4.559)	(4.526)	(4.565)	(4.505)	(4.527)
Senior High School	3.810	4.594	4.224	4.755	3.760	3.769
	(4.441)	(4.481)	(4.448)	(4.487)	(4.428)	(4.449)
Diploma 1/III	3.181	4.356	3.788	4.642	3.096	3.381
	(4.485)	(4.525)	(4.491)	(4.530)	(4.472)	(4.493)
Undergraduate	2.239	3.517	2.920	3.821	2.255	2.367
	(4.455)	(4.494)	(4.461)	(4.500)	(4.442)	(4.464)
Graduate/Doctoral	1.567	3.092	2.552	3.521	1.929	1.870
	(4.686)	(4.727)	(4.691)	(4.732)	(4.670)	(4.694)
Java	−2.522***	−2.210***	−2.356***	−2.065***	−2.536***	−2.139***
	(0.512)	(0.517)	(0.512)	(0.516)	(0.511)	(0.511)
Male	−0.793*	−1.044**	−0.817*	−1.183***	−0.796*	−0.794*
	(0.437)	(0.441)	(0.438)	(0.441)	(0.435)	(0.438)
Owner Age	0.0295	−0.00280	0.00816	−0.0118	0.0280	0.0196
	(0.0207)	(0.0207)	(0.0205)	(0.0206)	(0.0206)	(0.0207)
Firm Age	−0.0711*	−0.105***	−0.0804**	−0.113***	−0.0865**	−0.0694*
	(0.0396)	(0.0398)	(0.0396)	(0.0398)	(0.0393)	(0.0397)
Business License	−2.643***	−2.657***	−2.482***	−2.621***	−2.389***	−2.843***
	(0.519)	(0.524)	(0.520)	(0.525)	(0.518)	(0.521)
Legal Permit	6.187***	6.935***	6.106***	6.989***	6.284***	6.393***
	(0.685)	(0.687)	(0.688)	(0.688)	(0.682)	(0.685)
Ultra-micro	−5.165***	−7.145***	−6.125***	−7.397***	−4.952***	−6.113***
	(1.503)	(1.503)	(1.496)	(1.509)	(1.498)	(1.496)
Micro	−4.611***	−5.419***	−5.034***	−5.595***	−4.588***	−5.017***
	(1.400)	(1.411)	(1.401)	(1.413)	(1.395)	(1.401)
Sales (log)	0.434*	0.0672	0.275	0.0514	0.443*	0.331
	(0.252)	(0.251)	(0.250)	(0.253)	(0.250)	(0.251)
Elevation	0.000735	0.000743	0.000664	0.000717	0.000540	0.000658
	(0.000909)	(0.000917)	(0.000910)	(0.000918)	(0.000906)	(0.000910)
Internet Use	−2.653	−0.465	−0.937	0.504	−3.065	−1.204
	(2.490)	(2.509)	(2.479)	(2.499)	(2.483)	(2.482)
_cons	13.55**	21.33***	16.80**	21.65***	13.19*	14.84**
	(6.833)	(6.847)	(6.817)	(6.874)	(6.805)	(6.839)
R-squared	0.645	0.638	0.643	0.637	0.647	0.643
Number of Observation	5,016	5,016	5,016	5,016	5,016	5,016

Note: t statistics are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

In Table 6, we next investigate the impact of digital adoption on firm performance measured by sales growth. Our empirical results reveal that aggregate digital adoption is positively associated with sales growth, indicating that wider adoption is associated with higher performance. Considering the details, adopting e-procurement, digital payment, e-marketing, and e-commerce

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has a positive correlation with firm performance. Only POS adoption does not correlate with sales growth.

Table 7. Digital Adoption and Sales Growth (IV: Digital Competition)

	(1) Sales growth	(2) Sales growth	(3) Sales growth	(4) Sales growth	(5) Sales growth	(6) Sales growth
Aggregate Index	0.115***					
	(0.0353)					
Index E-Commerce		0.113***				
		(0.0356)				
Index E-Procurement			0.0750***			
			(0.0232)			
Index POS				0.396***		
				(0.147)		
Index Digital Payment					0.111***	
					(0.0342)	
Index E-Marketing						0.0999***
						(0.0308)
Transportaion	1.203	1.361	1.471	−1.778	1.273	1.397
	(1.080)	(1.096)	(1.080)	(1.768)	(1.076)	(1.079)
ICT	−4.517*	−4.150	−4.457*	−6.273**	−4.913**	−4.120
	(2.501)	(2.544)	(2.506)	(3.103)	(2.501)	(2.505)
Industry	1.813	1.864	1.982*	0.689	2.131*	1.543
	(1.156)	(1.174)	(1.154)	(1.498)	(1.147)	(1.169)
Trade	−0.890	−0.959	−1.111	−2.019	−0.632	−0.481
	(1.128)	(1.152)	(1.145)	(1.492)	(1.115)	(1.115)
Other Services	2.910***	3.345***	2.536**	1.439	3.248***	3.093***
	(1.104)	(1.101)	(1.138)	(1.549)	(1.083)	(1.095)
Elementary School	6.757	7.004	6.696	6.546	6.692	6.734
	(4.646)	(4.726)	(4.656)	(5.606)	(4.635)	(4.653)
Junior High School	3.270	3.433	3.197	3.494	3.203	3.225
	(4.518)	(4.596)	(4.528)	(5.450)	(4.507)	(4.525)
Senior High School	3.991	4.151	4.061	3.466	4.040	3.846
	(4.448)	(4.523)	(4.457)	(5.381)	(4.437)	(4.458)
Diploma 1/III	3.485	3.597	3.539	2.503	3.550	3.503
	(4.499)	(4.575)	(4.508)	(5.470)	(4.487)	(4.506)
Undergraduate	2.522	2.648	2.626	1.168	2.684	2.468
	(4.474)	(4.548)	(4.481)	(5.468)	(4.459)	(4.482)
Graduate/Doctoral	2.008	1.907	2.281	−0.825	2.445	2.056
	(4.708)	(4.793)	(4.710)	(5.882)	(4.685)	(4.714)
Java	−2.499***	−2.680***	−2.494***	−3.345***	−2.452***	−2.178***
	(0.530)	(0.559)	(0.531)	(0.790)	(0.525)	(0.512)
Male	−0.840*	−0.742	−0.699	−1.919***	−0.880**	−0.805*
	(0.448)	(0.463)	(0.460)	(0.594)	(0.444)	(0.451)
Owner Age	−0.0637	−0.0687*	−0.0600	−0.0116	−0.0825**	−0.0607
	(0.0414)	(0.0414)	(0.0420)	(0.0610)	(0.0390)	(0.0419)
Firm Age	−2.594***	−2.723***	−2.414***	−2.951***	−2.413***	−2.795***
	(0.518)	(0.528)	(0.524)	(0.636)	(0.521)	(0.521)
Business License	6.338***	6.793***	5.837***	6.476***	6.482***	6.443***
	(0.710)	(0.696)	(0.770)	(0.844)	(0.697)	(0.703)
Legal Permit	−5.640***	−6.218***	−5.726***	−0.539	−5.687***	−6.262***
	(1.593)	(1.564)	(1.588)	(3.153)	(1.584)	(1.537)
Ultra-micro	−4.789***	−4.851***	−4.842***	−2.937	−4.868***	−5.060***
	(1.423)	(1.444)	(1.423)	(1.966)	(1.415)	(1.411)
Micro	0.347	0.161	0.342	1.316**	0.319	0.297
	(0.266)	(0.256)	(0.267)	(0.564)	(0.263)	(0.262)
Sales (log)	0.000733	0.000856	0.000657	0.00152	0.000584	0.000664
	(0.000909)	(0.000926)	(0.000911)	(0.00114)	(0.000907)	(0.000910)
Elevation	−2.090	−3.133	−1.427	−5.960	−2.092	−1.080
	(2.611)	(2.780)	(2.558)	(3.864)	(2.605)	(2.531)
Internet Use	16.00**	20.26***	15.78**	0.836	16.30**	16.09**
	(6.941)	(6.852)	(6.973)	(11.20)	(6.902)	(6.944)
R-squared	0.644	0.632	0.643	0.482	0.646	0.643
Number of Observation	5,016	5,016	5,016	5,016	5,016	5,016

Note: t statistics are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

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In Table 7, we continue to analyze the impact of digital adoption on business performance by examining digital competition, which is measured by the availability of online payment for surrounding firms as the IV. The result provides uniform evidence that aggregate adoption and all subindices of digital adoption are associated with increased sales growth for UMSEs, confirming the findings of several previous studies (Falentina et al., 2022).

Table 8. Ordinary Least Squares: Digital Adoption and Financial Literacy

	(1)	(2)	(3)	(4)	(5)	(6)
	FinLitIndex	FinLitIndex	FinLitIndex	FinLitIndex	FinLitIndex	FinLitIndex
Aggregate Index	0.000654***					
	(0.000174)					
Index E-Commerce		0.000365***				
		(0.000113)				
Index E-Procurement			0.0000447			
			(0.0000793)			
Index POS				0.000334***		
				(0.000109)		
Index Digital Payment					0.000885***	
					(0.000169)	
Index E-Marketing						0.000285*
						(0.000149)
Transportation	−0.0190	−0.0178	−0.0174	−0.0202	−0.0190	−0.0176
	(0.0142)	(0.0142)	(0.0142)	(0.0142)	(0.0141)	(0.0142)
ICT	0.0799**	0.0812**	0.0801**	0.0789**	0.0765**	0.0811**
	(0.0329)	(0.0329)	(0.0329)	(0.0329)	(0.0329)	(0.0329)
Industry	−0.0449***	−0.0437***	−0.0425***	−0.0438***	−0.0434***	−0.0444***
	(0.0151)	(0.0151)	(0.0151)	(0.0151)	(0.0151)	(0.0152)
Trade	0.0229	0.0240	0.0255*	0.0247*	0.0236	0.0256*
	(0.0147)	(0.0146)	(0.0147)	(0.0146)	(0.0146)	(0.0146)
Other Services	−0.0815***	−0.0782***	−0.0778***	−0.0790***	−0.0806***	−0.0788***
	(0.0142)	(0.0142)	(0.0142)	(0.0142)	(0.0141)	(0.0142)
Elementary School	−0.0233	−0.0211	−0.0202	−0.0213	−0.0247	−0.0215
	(0.0612)	(0.0612)	(0.0613)	(0.0612)	(0.0611)	(0.0612)
Junior High School	−0.0267	−0.0252	−0.0245	−0.0249	−0.0279	−0.0255
	(0.0594)	(0.0594)	(0.0595)	(0.0595)	(0.0593)	(0.0595)
Senior High School	−0.000817	0.00172	0.00340	0.00266	−0.00222	0.00111
	(0.0584)	(0.0584)	(0.0585)	(0.0584)	(0.0583)	(0.0585)
Diploma 1/III	0.0268	0.0303	0.0333	0.0320	0.0246	0.0305
	(0.0590)	(0.0590)	(0.0591)	(0.0590)	(0.0589)	(0.0590)
Undergraduate	0.0213	0.0252	0.0284	0.0269	0.0196	0.0251
	(0.0586)	(0.0586)	(0.0587)	(0.0586)	(0.0585)	(0.0587)
Graduate/Doctoral	0.0414	0.0453	0.0504	0.0471	0.0412	0.0465
	(0.0616)	(0.0616)	(0.0617)	(0.0616)	(0.0615)	(0.0617)
Java	0.0341***	0.0345***	0.0361***	0.0354***	0.0334***	0.0361***
	(0.00674)	(0.00674)	(0.00674)	(0.00672)	(0.00673)	(0.00672)
Male	−0.00323	−0.00369	−0.00473	−0.00568	−0.00283	−0.00401
	(0.00575)	(0.00575)	(0.00576)	(0.00574)	(0.00574)	(0.00576)
Owner Age	−0.00196***	−0.00205***	−0.00215***	−0.00208***	−0.00192***	−0.00208***
	(0.000273)	(0.000270)	(0.000269)	(0.000269)	(0.000271)	(0.000271)
Firm Age	0.00144***	0.00133**	0.00126**	0.00127**	0.00140***	0.00135***
	(0.000520)	(0.000519)	(0.000520)	(0.000518)	(0.000518)	(0.000522)

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Business License	0.0403***	0.0399***	0.0405***	0.0400***	0.0417***	0.0398***
	(0.00683)	(0.00684)	(0.00684)	(0.00684)	(0.00683)	(0.00685)
Legal Permit	−0.0467***	−0.0436***	−0.0436***	−0.0434***	−0.0471***	−0.0445***
	(0.00901)	(0.00896)	(0.00905)	(0.00896)	(0.00898)	(0.00900)
Ultra-micro	0.0716***	0.0647***	0.0612***	0.0663***	0.0755***	0.0640***
	(0.0198)	(0.0196)	(0.0197)	(0.0196)	(0.0197)	(0.0197)
Micro	0.0759***	0.0735***	0.0714***	0.0732***	0.0772***	0.0726***
	(0.0184)	(0.0184)	(0.0184)	(0.0184)	(0.0184)	(0.0184)
Sales (Log)	0.00649**	0.00504	0.00471	0.00567*	0.00701**	0.00535
	(0.00331)	(0.00327)	(0.00329)	(0.00329)	(0.00330)	(0.00330)
Elevation	0.0000263***	0.0000266**	0.0000261**	0.0000268**	0.0000252**	0.0000260**
	(0.0000119)	(0.0000120)	(0.0000120)	(0.0000120)	(0.0000119)	(0.0000120)
Internet Use	0.0594*	0.0630*	0.0740**	0.0696**	0.0533	0.0702**
	(0.0327)	(0.0327)	(0.0326)	(0.0325)	(0.0327)	(0.0326)
_cons	0.539***	0.571***	0.576***	0.559***	0.527***	0.561***
	(0.0899)	(0.0893)	(0.0896)	(0.0895)	(0.0896)	(0.0899)
R-squared	0.934	0.934	0.934	0.934	0.934	0.934
Number of Observation	5,016	5,016	5,016	5,016	5,016	5,016

Note: t statistics are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

We also estimate the impact of digital adoption on the level of financial literacy of UMSEs' owners to answer the final research question. Subsequently, we employ 2SLS estimations (Table 9) using signal strength as an IV, and this method is basically to tackle potential bias from the endogeneity issue of business performance and financial aspects. Our results show that digital adoption is positive and significantly associated with financial literacy, suggesting that digital adoption increases among UMSEs correspond to an increase in their financial knowledge as well. In turn, this will boost their business productivity as a whole.

Table 9. Two-stage Least Squares: Digital Adoption and Financial Literacy (IV: Signal Strength)

	(1) FinLiteracyIndex	(2) FinLiteracyIndex	(3) FinLiteracyIndex	(4) FinLiteracyIndex	(5) FinLiteracyIndex	(6) FinLiteracyIndex
Aggregate Index	0.007***					
	(0.00220)					
Index E-Commerce		0.007***				
		(0.00258)				
Index E-Procurement			0.0182			
			(0.0153)			
Index POS				0.007***		
				(0.00236)		
Index Digital Payment					0.006***	
					(0.00175)	
Index E-Marketing						0.006***
						(0.001)
Transportation	−0.036**	−0.025	−0.018	−0.077***	−0.029*	−0.0222
	(0.0172)	(0.0194)	(0.0481)	(0.0274)	(0.0159)	(0.0162)
ICT	0.0780**	0.104**	0.0789	0.0534	0.0539	0.101***
	(0.0380)	(0.0454)	(0.112)	(0.0448)	(0.0369)	(0.0379)
Industry	−0.0729***	−0.0698***	−0.109	−0.0725***	−0.0497***	−0.0855***

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	(0.0195)	(0.0225)	(0.0755)	(0.0225)	(0.0168)	(0.0210)
Trade	−0.0107	−0.0152	−0.164	−0.00122	0.00884	0.0173
	(0.0198)	(0.0242)	(0.166)	(0.0215)	(0.0168)	(0.0168)
Other Services	−0.129***	−0.1000***	−0.357	−0.117***	−0.103***	−0.113***
	(0.0221)	(0.0207)	(0.239)	(0.0230)	(0.0170)	(0.0190)
Elementary School	−0.0612	−0.0466	−0.161	−0.0503	−0.0551	−0.0545
	(0.0715)	(0.0836)	(0.239)	(0.0823)	(0.0681)	(0.0704)
Junior High School	−0.0530	−0.0431	−0.132	−0.0369	−0.0504	−0.0504
	(0.0690)	(0.0810)	(0.221)	(0.0795)	(0.0658)	(0.0681)
Senior High School	−0.0514	−0.0409	−0.176	−0.0215	−0.0403	−0.0539
	(0.0692)	(0.0807)	(0.249)	(0.0784)	(0.0654)	(0.0685)
Diploma 1/III	−0.0516	−0.0448	−0.256	−0.00948	−0.0347	−0.0405
	(0.0722)	(0.0843)	(0.315)	(0.0800)	(0.0676)	(0.0704)
Undergraduate	−0.0638	−0.0557	−0.279	−0.0207	−0.0408	−0.0568
	(0.0725)	(0.0845)	(0.325)	(0.0799)	(0.0672)	(0.0710)
Graduate/Doctoral	−0.0655	−0.0736	−0.294	−0.0367	−0.0220	−0.0486
	(0.0784)	(0.0935)	(0.356)	(0.0872)	(0.0707)	(0.0756)
Java	0.00914	−0.00286	−0.0645	0.0169	0.0151	0.0308***
	(0.0109)	(0.0160)	(0.0873)	(0.0110)	(0.00938)	(0.00781)
Male	0.0160*	0.0227*	0.106	−0.0198**	0.0108	0.0159*
	(0.00890)	(0.0121)	(0.0945)	(0.00906)	(0.00763)	(0.00879)
Owner Age	0.000303	0.000468	0.00498	−0.000296	−0.000356	−0.000251
	(0.000767)	(0.000954)	(0.00605)	(0.000712)	(0.000573)	(0.000619)
Firm Age	0.00366***	0.00323***	0.0120	0.00203***	0.00242***	0.00376***
	(0.00911)	(0.000969)	(0.00915)	(0.000739)	(0.000654)	(0.000923)
Business License	0.0385***	0.0295***	0.0808**	0.0311***	0.0499***	0.0272***
	(0.00790)	(0.00998)	(0.0409)	(0.00962)	(0.00795)	(0.00863)
Legal Permit	−0.0889***	−0.0583***	−0.329	−0.0535***	−0.0737***	−0.0774***
	(0.0167)	(0.0132)	(0.241)	(0.0125)	(0.0129)	(0.0141)
Ultra-micro	0.196***	0.158***	0.521	0.193***	0.172***	0.143***
	(0.0447)	(0.0421)	(0.391)	(0.0509)	(0.0373)	(0.0321)
Micro	0.130***	0.126***	0.270	0.121***	0.117***	0.107***
	(0.0270)	(0.0311)	(0.178)	(0.0295)	(0.0238)	(0.0233)
Sales (log)	0.0279***	0.0156***	0.0861	0.0293***	0.0227***	0.0222***
	(0.00765)	(0.00578)	(0.0691)	(0.00925)	(0.00610)	(0.00619)
Elevation	0.0282**	0.0365**	0.0154	0.0407**	0.0193	0.0238*
	(0.0138)	(0.0166)	(0.0415)	(0.0167)	(0.0133)	(0.0136)
Internet Use	−0.112*	−0.184*	−0.434	−0.0456	−0.0847	−0.0304
	(0.0650)	(0.0973)	(0.440)	(0.0588)	(0.0562)	(0.0474)
_cons	0.0897	0.370***	−1.157	0.120	0.192	0.162
	(0.173)	(0.140)	(1.484)	(0.193)	(0.144)	(0.155)
R-squared	0.912	0.878	0.237	0.882	0.920	0.914
Number of Observation	5,016	5,016	5,016	5,016	5,016	5,016

Note: t statistics in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

5 Conclusion

This study examines digital adoption by UMSEs in Indonesia by employing a survey of 5,035 UMSEs in 17 major provinces. We construct an index to measure the digital adoption index by identifying UMSE business processes that could potentially be transformed with digital technology, considering five channels of e-procurement, POS, digital payment, e-marketing, and e-commerce. Our digital adoption index identifies whether firms adopted digital technology and considers the usage dimension to measure the level of adoption more precisely. The average digital adoption index is 20.33 out of 100. E-procurement has the highest level of adoption, while POS has the lowest. A significant gap is evident in the digital adoption between UMSEs in and outside Java Island.

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Moreover, digital adoption by small enterprises is higher than that by ultra-micro and micro enterprises.

The digital adoption index enables us to investigate the determining factors for explaining the disparity in digital adoption by UMSEs in Indonesia. Our results reveal that firm-specific factors, owner demographic characteristics, business environment, ICT infrastructure quality, and culture explain the disparity in UMSEs' digital adoption. Going deeper, we also find evidence that digital adoption significantly improves UMSEs' performance. Finally, digital adoption contributes to enhancing UMSEs' level of financial literacy.

Several policy implications emerge from our results. We provide strong evidence that digital adoption is beneficial for improving UMSEs performance. Digital adoption is also found to contribute to UMSEs' financial inclusion and UMSE owners' financial literacy. Notable, the level of digital adoption of UMSEs in Indonesia remains relatively low. While UMSEs may have adopted some digital technologies, their use is relatively limited; therefore, policies and initiatives to encourage, assist, and facilitate UMSEs to digitally transform business processes should be intensively and continuously implemented by governmental organizations, financial services providers, nongovernmental organizations, the academic community, and other stakeholders. Our results also demonstrate that technology awareness and trust in digital processes are important factors of digital adoption by UMSEs in Indonesia; therefore, concerted efforts to improve digital literacy on a broad scale are essential. Most of the UMSEs in Indonesia do not have legal entities and business permits, and most do not separate accounting from the owner's account. These factors are found to have a significant effect on digital adoption. Therefore, more flexible regulations on legal entities and business permits should be applied to UMSEs.

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Appendix

Table A1. Correlation Matrix

	Aggregate Index	Index E-commerce	Index E-Procurement	Index POS	Index Digital Payment	Index E-Marketing	Fin. Literacy Index	Fin. Inclusion Index	Sales Growth	Firm Age	Business License
Aggregate Index	1.000										
Index E-commerce	0.705	1.000									
Index E-Procurement	0.799	0.408	1.000								
Index POS	0.524	0.197	0.184	1.000							
Index Digital Payment	0.732	0.462	0.489	0.267	1.000						
Index E-Marketing	0.709	0.416	0.479	0.196	0.507	1.000					
Fin. Literacy Index	0.077	0.089	0.016	0.057	0.097	0.038	1.000				
Fin. Inclusion Index	0.441	0.296	0.298	0.169	0.491	0.362	0.219	1.000			
Sales Growth	0.166	0.059	0.163	0.028	0.179	0.161	−0.153	0.068	1.000		
Firm Age	−0.195	−0.136	−0.150	−0.061	−0.136	−0.206	−0.026	−0.095	−0.035	1.000	
Business License	0.080	0.027	0.057	0.050	0.019	0.132	0.037	0.115	0.021	0.005	1.000
Home-Based Location	0.102	0.050	0.059	0.106	0.103	0.049	−0.050	0.036	0.064	0.089	−0.036
Owners Age	−0.243	−0.196	−0.163	−0.100	−0.205	−0.209	−0.141	−0.165	0.003	0.389	0.133
Dummy Male	−0.074	−0.077	−0.073	0.040	−0.058	−0.092	−0.022	−0.025	−0.024	0.069	−0.010
Education	0.250	0.169	0.178	0.118	0.201	0.227	0.085	0.216	0.010	−0.146	0.118
Digital Competition	0.444	0.310	0.327	0.120	0.426	0.425	0.150	0.366	0.050	−0.151	0.186
Banking Relationship	0.035	−0.004	0.023	0.060	0.023	0.018	0.018	0.021	0.023	0.152	0.146
Smartphone Society	0.125	0.086	0.093	0.037	0.099	0.134	0.071	0.099	−0.027	−0.021	0.026
Technology Awareness	0.193	0.085	0.162	0.094	0.193	0.151	0.034	0.145	0.228	−0.009	−0.134
Trust to Digital	0.232	0.090	0.186	0.084	0.202	0.275	0.032	0.148	0.036	−0.073	0.006
Log Sales	0.125	0.080	0.064	0.121	0.131	0.061	0.029	0.137	0.112	0.134	0.163
Log GDP	0.120	0.128	0.056	0.040	0.141	0.087	−0.154	0.008	0.078	−0.010	−0.049
Dummy Java	0.115	0.117	0.084	0.057	0.110	0.033	0.120	0.135	−0.065	−0.067	−0.038
Internet Use	0.101	0.109	0.629	0.041	0.098	0.051	0.019	0.037	−0.009	−0.013	−0.159
Elevation	−0.029	−0.042	−0.016	−0.027	−0.009	−0.005	0.031	−0.042	0.007	0.001	0.069

Continued

	Home-Based Location	Owners Age	Gender Male	Education	Digital Competition	Banking Relationship	Smartphone Society	Technology Awareness	Trust to Digital	Log Sales	Log GDP	Dummy Java	Internet Use	Elevation
Home-Based Location	1.000													
Owners Age	0.026	1.000												
Dummy Male	0.139	0.053	1.000											
Education	0.061	−0.119	0.003	1.000										
Digital Competition	0.049	−0.163	−0.067	0.173	1.000									
Banking Relationship	0.064	0.175	0.028	0.005	0.055	1.000								
Smartphone Society	0.024	−0.020	−0.006	0.051	0.177	0.034	1.000							
Technology Awareness	0.152	−0.067	0.013	0.131	0.094	−0.021	0.085	1.000						
Trust to Digital	0.047	−0.074	−0.028	0.162	0.149	0.029	0.079	0.232	1.000					
Log Sales	0.181	0.085	0.145	0.096	0.110	0.173	0.043	0.142	0.004	1.000				
Log GDP	0.116	0.029	0.054	−0.006	0.048	0.009	0.030	0.084	−0.115	0.162	1.000			
Dummy Java	0.020	−0.141	0.023	0.030	0.189	0.052	0.021	−0.007	0.014	0.067	−0.117	1.000		
Internet Use	0.039	−0.055	−0.031	0.030	0.072	−0.049	0.063	0.093	−0.011	0.036	0.234	0.025	1.000	
Elevation	−0.045	0.012	0.004	0.013	0.009	0.016	−0.007	−0.042	−0.003	−0.058	−0.219	0.098	−0.407	1.000

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